

Project name: Cornetto: Combinatorial and Relational Network as Toolkit for Dutch Language Technology

Project number: STE05039

Reporting period: 1 April 2008 – 30 June 2008

Participants: Vrije Universiteit, Amsterdam, Faculteit der Letteren (VU-AMS)
Universiteit van Amsterdam, Instituut voor Informatica (UVA-AMS)
K.U. Leuven- ICRI (KU-LEU)
Irion Technologies B.V. (IRION)

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Summary

1.1 Deliverables

The next table lists the deliverables with related work packages, the date of delivery and the status. The numbering is in accordance with appendix-4 of the consortium agreement.

Table 1: Deliverables

Id	Wp	Title	delivery date	status	Mo.
	0	project management en rapportering	continuing	done	
D01	1	Cornetto Database	30/09/06	done	6
D02	2	Aligned RBN, DWN and Wordnet2.x	30/09/06	done	6
D03	3	Top-level ontology, relation constraints and assignments	31/12/06	done	9
D04	4	Core Cornetto	30/07/07	done	15
D05	4	Extended Cornetto	31/01/08	done	22
D06	5	Acquired concepts	30/03/07	done	12
D07	5	Acquired relations	30/07/07	done	15
D08	5	Cornetto Acquisition toolkit	30/07/07	done	15
D09	6	Ontology for the legal and finance domain	30/03/07	done	12
D10	6	Sub-database tuned, trimmed and extended to the legal and finance domain	31/01/08	done	22
D11	6	Domain tuning toolkit	31/01/08	done	22
D12	7	Evaluation report of the ontology advisory-board	30/03/07	WORKSHOP	12
D13	7	Task-based evaluation of Core Cornetto	30/09/07	done	17
D14	7	Task-based evaluation of Extended Cornetto	31/01/08	done	22
D15	7	Evaluation report of the user-group on the legal and finance sub-database	31/01/08	EXTERNAL	22
D16	8	Final Cornetto database and software	30/03/08	done	24

1.2 Finished deliverables

All deliverables have been finished. Deliverable D15 has been replaced by external evaluation of the final data by Polderland BV. This will be carried out in the period October-December 2008 for which a budget is reserved of 5,000 euros.

1.3 Changes

All activities have been completed and the data have been delivered to the TST-centrale on October 2nd, 2008.

1.4 Spent effort compared to planned effort

partner	Total person months whole project	Nr. person months spent for period 5	Nr. person months spent in total
VU-AMS	15 (18)*	0 (2.9)	15(24.3)+
UVA-AMS	15	0(0.38)	15(18.28)+
KU-LEU	14	5 (5.33)	10.53 (10.53)-
IRION	13 (10)*	0 (1)	13(17.8)+
Totaal	57		

Comment: the straight number indicate the paid effort in person months, the numbers between brackets give the actual person months including the work exceeding the paid effort.

+ Overspending is based on the overhead of the organization and voluntary contributions.

* Due to the budget shift from VUA to Irion Technologies, the totals have been adjusted.

- Due to higher personnel costs, the total number of person months by K.U.Leuven is less than was foreseen in the proposal, where 14 person months were foreseen.

1.5 Dissemination of the results Cornetto

Papers

Period 4

Valentin Jijkoun and Katja Hofmann: "Generating a Non-English Subjectivity Lexicon: Relations That Matter". Ingediend bij EACL 2009.

Erik Tjong Kim Sang and Katja Hofmann: "Lexical Patterns or Dependency Patterns: Which Is Better for Hypernym Extraction?". Ingediend bij EACL 2009.

Boiy, E., Deschacht, K. & Moens, M.-F. (2008). Learning Visual Entities and their Visual Attributes from Text Corpora In *Proceedings of the 5th International Workshop on Text-based Information Retrieval*. IEEE Press..

A. Horák, I. Maks, A. Rambousek, R. Segers, H. van der Vliet, P. Vossen (fc) "[Cornetto Tools and Methodology for Interlinking Lexical Units, Synsets and Ontology](#)", in: Proceedings of the 18th International Congress of Linguists ([CIL18](#)), Seoul, Republic of Korea, July 21-26, 2008.

A. Pease, C. Fellbaum, P. Vossen, (fc) "[Building the Global WordNet Grid](#)", in: Proceedings of the 18th International Congress of Linguists ([CIL18](#)), Seoul, Republic of Korea, July 21-26, 2008.

Vossen P., Fellbaum C. (fc) "Universals and Idiosyncracies in Multilingual WordNets", in: Handbook Multilingual Lexicography, [Oxford University Press](#), 2008

[Vossen P.](#), [Maks I.](#), [Segers R.](#), [VanderVliet H.](#) (fc) "[Integrating lexical units, synsets and ontology in the Cornetto Database](#)", in: Proceedings of [LREC 2008](#), Marrakech, Morocco, May 28-30 May 2008.

[Maks I.](#), [P. Vossen](#), [Segers R.](#), [VanderVliet H.](#), [van Zutphen H.](#) (fc) "Encoding adjectives in the Dutch semantic lexical database Cornetto", in: Proceedings of [LREC 2008](#), Marrakech, Morocco, May 28-30 May 2008.

Erik Tjong Kim Sang and Katja Hofmann, Automatic Extraction of Dutch, Hypernym-Hyponym Pairs. In Proceedings of CLIN-2006, Leuven, Belgium, 2007.

[Horak A.](#), [P. Vossen](#), [A. Rambousek](#) "[A Distributed Database System for Developing Ontological and Lexical Resources in Harmony](#)", in: Proceedings of the 9th International Conference on Intelligent Text Processing and Computational Linguistics: [CICLing 2008](#), February 17-23, 2008, Haifa, Israel. Also to be published in the [Lecture Notes on](#)

Computational Linguistics and Intelligent Text Processing in [Lectures Notes in Computer Science](#), Volume 4919/2008, ISBN 978-3-540-78134-9, 1-15, Springer-Verlag, Berlin, 2008.

Horak A., Vossen P., Rambousek A. (2008) The Development of a Complex-Structured Lexicon based on WordNet, in: Proceedings of the Fourth International GlobalWordNet Conference - [GWC 2008](#), Szeged, Hungary, January 22-25, 2008

Vossen P., Maks I., Segers R., VanderVliet H., van Zutphen H. (2008) "The Cornetto Database: the architecture and alignment issues", in: Proceedings of the Fourth International GlobalWordNet Conference - [GWC 2008](#), Szeged, Hungary, January 22-25, 2008

Period 3

Vossen P. and C. Fellbaum, 2007 (fc) "Universals and Idiosyncracies in Multilingual WordNets", In: Handbook Multilingual Lexicography, [Oxford University Press](#), 2007

Period 2

Fellbaum C. and P. Vossen, 2007 "Connecting the Universal to the Specific: Towards the Global Grid", In: Proceedings of [The First International Workshop on Intercultural Collaboration](#) (IWIC 2007), Kyoto, Japan, January 25-26, 2007 Publically available on external Website

Vossen, P. (2006). *Cornetto: Een lexicaal-semantische database voor taaltechnologie*, Dixit Special Issue, Stevin. ([PDF](#)) Publically available on external Website

Vossen, P. (2006). *Een communicatief lexicon door het verankeren van woordbetekenissen*, Oratie. ([PDF](#)) Publically available on external Website

Posters

Period 3

Vossen, P., Hofmann, K., de Rijke, M., Tjong Kim Sang, E., and Deschacht, K. (2007). The Cornetto Database: Architecture and User-Scenarios. In DIR 2007, pp. 89-96. ([PDF](#)) Publically available on external Website

Tjong Kim Sang, E. (2007) Extracting Hypernym Pairs from the Web. In Proceedings of ACL 2007 (poster), Prague, Czech Republic.

Period 2

Hofmann, K. and Tjong Kim Sang, E. (2007). Automatic Extension of Non-English WordNets. SigIR 2007, accepted. Publically available on external Website

Talks

Period 5

Poster presentation by Piek Vossen on Stevin-dag, Hoeven, 2008.

Presentation by Roxane Segers at the Stevin-Imix evaluation workshop, May 2008, Rotterdam.

Period 4

Guest Lecture on [Corpusgebaseerd tekstonderzoek](#) voor het [ICT](#) Onderwijscentrum van de Vrije Universiteit Amsterdam, December 13, 2007.

Guest Lecture on Corpus-based Methods: "[Automatic term extraction from domain corpora](#)", Universiteit Nijmegen, November 26, 2007

Invited speaker on [Taal, Intelligentie en Betekenis](#)" on the Lustrum symposium [Kunstmatige Intelligentie, Ontwikkelingen en Toepassingen in Muziek, Taal en Geneeskunde](#)" van het [Genootschap Physica](#), Alkmaar, October 27, 2007

Invited Speaker on [Global Wordnet Grid](#) on the 6th International [PLAIN](#) Language Conference, Amsterdam, Netherlands, October 11-14, 2007,

Invited speaker on [Global Wordnet Grid](#) at the [Politechnika Koszalin](#), Koszalin, Poland, October 8, 2007.

Keynote Speaker on [Global WordNet](#) at the [3rd Language & Technology Conference](#): Human Language Technologies as a Challenge for Computer Science and Linguistics, Poznan, Poland, October 5-7, 2007,

Period 3

Tjong Kim Sang, E.: Cornetto presentation on the [Stevin](#) Programmadag, Nederlandse TaalUnie, Hoeven, September 21, Publically available on external Website

Vossen, P, Guest lecture: "Cornetto: Een Combinatorische Semantische Database voor het Nederlands" for the [Capita Selecta](#) of the Instituut voor Nederlandse Lexicologie ([INL](#)), Leiden, May 22, 2007. Publically available on external Website

Vossen, P, Guest lecture: Global WordNet and Global WordNet Grid at the [University of Potchefstroom](#), South Africa, March 19, 2007. Publically available on external Website

Vossen, P, Guest lecture: Cornetto and Wordnets at the [University of Potchefstroom](#), South Africa, March 20, 2007. Publically available on external Website

Vossen, P, Guest lecture: "African Languages WordNet Workshop/Training" of the [Department African Languages](#) of the University of South Africa, Unisa), Pretoria, South Africa, March 21-23, 2007.

Period 2

Tjong Kim Sang, E.: "Extracting Dutch Hypernymy Pairs from the Web". Presented at CLIN-17, 12 January 2007 in Leuven, Belgium. Publically available on external Website

Vossen, P., Maks, I., Martin, W., van der Vliet, H., Hofmann, K., van Zutphen, H.: *The Cornetto Database*. Presented at CLIN-17, 12 January 2007 in Leuven. ([Abstract](#), [PDF](#), [PowerPoint](#)) Publically available on external Website

Vossen, P.: *Een communicatief lexicon door het verankeren van woordbetekenissen*, Oratie. Presented December 22 2006 at Vrije Universiteit, Amsterdam. ([PDF](#), [PowerPoint](#)) Publically available on external Website

Vossen, P.: *Er kan meer dan men doet met de Cornetto database*. Presented November 30 2006 at the TST Themadag: de gebruiker centraal in Rotterdam. ([PDF](#), [PowerPoint](#)) Publically available on external Website

Period 1

Vossen, P.: *Cornetto*. Presented September 11 2006 at the Stevin project day in Antwerp. ([PDF](#), [PowerPoint](#)) Publically available on external Website

2 Progress per deliverable

2.1 Realized work in the reporting period

WP0: Management

In the 5th period, we had no project meetings.

WP1: Cornetto database

Completed

WP2: Aligning semantic resources

Completed

WP3: Ontologize

Completed

WP4: Cornetto editing

The final data have been edited and quality and status labels have been derived for all delivered synsets. We have carried out various final consistency checking tests for which software was developed. We delivered more data than promised, differentiated in status so that future users have maximum recall and can differentiate the quality of the data.

WP5: Acquisition toolkit

Completed.

WP6: Domain acquisition

We implemented a complete toolkit which aids the lexicographer in creating a lexicon. The toolkit offers various functions and parameters to suit different lexicographical needs. In our experiments, we focused on two tasks: the extraction of domain-specific terms and the extraction of collocations. We have used the toolkit to create two domain-specific lexica (in the medical and legal domain) for the Dutch language. The methods are purely statistical in nature and learn the lexicographical knowledge from corpora. Software, documentation and reports with regard to the toolkit have been delivered.

WP7: Evaluation

Irion implemented 3 methods to implement semantic relations in the classification software:

1. Adding any content word combination (bi-grams) to the index and the classification text;
2. Only create combinations based on the Cornetto database in a window of 15 words;
3. Combined method 1 and 2;

Irion has a Dutch classification system that assigns thesaurus labels from the IPTC thesaurus (<http://www.iptc.org/pages/index.php>) to Dutch news text. We will investigate whether the above methods improve the current system on a test set that is defined for Irion's customers. This work was carried out and the report was delivered.

UvA evaluated Cornetto on the task of generating a subjectivity lexicon for Dutch. Our method is based on an English subjectivity lexicon, an online translation service, and a wordnet. The wordnet is viewed as a graph, which is initialized according to the automatic translations from the English lexicon. The weights are then propagated through the network using an iterative algorithm similar to PageRank. The result is a list of words ranked according to their subjectivity.

In this task we compared two wordnets: the 2001 version of the Dutch WordNet, and the 2008 version of Cornetto. Evaluation is based on human assessments of the top 2000 negative words and the top 1500 positive words. We find that the use of Cornetto results in a 7% improvement in accuracy. Between 70% and 86% of this improvement can be attributed to the larger size of Cornetto, the remaining improvement is attributed to the larger set of relations between words.

WP8: Distribution

More users have requested a license and received the data. The final data with the documentation has been delivered to the TST centrale.

2.2 Problems and solutions

No further problems have been reported

2.3 Suggested planning for the next period

The work has been completed. We agreed on evaluation of the data by Polderland BV. This will be carried out in the period October-December 2008.